ENVIRONMENTAL PERFORMANCE, LCA AND ESG SUSTAINABILITY REPORT

Project: **Example - tutorial house in Cape Town (ERA5 1991-2020) [costs to be entered by the user]**

Life Cycle Assessment (LCA), Carbon Footprint and Sustainable Certifications

⚠ **The grid emission factor was not informed: the emissions from pumping electricity are not meaningful until it is entered.**

⚠ **Investment, maintenance and electricity costs were not informed: the economic result shows only the savings on the water bill; payback and NPV are not meaningful until costs are entered.**

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Triton 0.1.0 · International Edition (INT)

ISO 14040 / ISO 14044 · GHG Protocol

CLIMATE LOCATION	CLIMATE BASIS	SERIES PERIOD	TARIFF LOCATION
Cape Town / South Africa	Observed series	01/01/1991 – 31/12/2020	-

TANK / SAVINGS	EMBODIED CARBON	AVOIDED CO2 / YR	20-YEAR NET CARBON	CARBON PAYBACK
1,000 L	**160.4 kg**	**+0.1 kg/yr**	**+162 kg**	**> 50 years**
9.5% potable replaced	Materials & Equipment	Net Annual Balance	Lifetime Net Mitigation	-0.39 g CO2e/L avoided

1. 1. BUILDING & DEMAND INPUTS

Parameter	Value
Rainfall Series	**CapeTown.csv**
Catchment Area (Roof)	**150.0 m²**
Runoff Coefficient (C)	0.85
First Flush Discard	2.0 mm
Occupants	4 occupants
Per-capita Demand	150 L/person/day
Rainwater Target Share	**28%**

2. 2. HYDRAULIC BALANCE (YAS MODEL)

Hydraulic Indicator	Result
Storage Tank Volume	**1,000 litres**
Potable Savings	**9.5%**
Annual Pluvial Use	**20.8 m³/yr (57.1 L/day)**
Residual Potable Demand	198.3 m³/yr (542.9 L/day)
Annual Overflow	23.8 m³/yr (65.2 L/day)
Full Reliability Days	**29.7% of days**
Total Water Demand	219.2 m³/yr

3. 3. LIFE CYCLE INVENTORY (LCI — ISO 14040/14044)

Component / Phase	Quantity / Size	Embodied CO2	Op. CO2/Yr	20-Yr Balance	Source / Reference
Lower Cistern (Polyethylene)	1,000 L (22.0 kg)	**+35.5 kg**	--	**+35.5 kg**	Resin: Franklin Associates for the American Chemistry Council (out/2020), 'Cradle-to-Gate Life Cycle Analysis of High-Density Polyethylene (HDPE) Resin', Table 6: 1,612 kg CO2e per 1,000 kg of resin (TRACI 2.1 / IPCC 2013 GWP-100, US scope). Replaces the previous internal value of 3.00, which had no source and declared the same boundary. The mass per m^3 remains an internal scenario.
Rigid PVC Pipe (DN 32 mm)	20.0 m (0.71 kg CO2/m)	**+14.1 kg**	--	**+14.1 kg**	Manufacturer's data sheet (2025), solvent-weld pipe 32 x 6 m: 1,793.14 g × 2.36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3)
Rigid PVC Pipe (DN 40 mm)	5.0 m (1.01 kg CO2/m)	**+5.0 kg**	--	**+5.0 kg**	Manufacturer's data sheet (2025), solvent-weld pipe 40 x 6 m: 2,561.27 g × 2.36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3)
Rigid PVC Pipe (DN 100 mm)	20.0 m (2.78 kg CO2/m)	**+55.5 kg**	--	**+55.5 kg**	Standardised pipe geometry, reinforced series: OD 101.6 mm, wall 2.5 mm × 2.36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3)
Pressurising Pump (0.24 CV - Cast Iron, Copper & Aluminium)	1 pcs (main) — 0.24 CV	**+29.1 kg**	--	**+29.1 kg**	Manufacturer catalogue mass (60 Hz) × 4.77 kg CO2e/kg from the Grundfos EPD EPD-GRU-20250340-CBA1-EN (A1-A3): order-of-magnitude approximation; the EPD of the manufacturer actually used prevails
First-Flush Discard System (Leaf Filter + Tank 300 L)	300 L (9.3 kg material)	**+21.1 kg**	--	**+21.1 kg**	First-flush discard / Polymer & Stainless-steel inventory
Pumping Electricity Use	21.2 kWh/yr (1.02 kWh/m^3)	**--**	--	--	Editable grid factor; use the factor of the site and of the study period
Treatment Line	20.8 m^3/yr; 0,048 kg product/yr (1.50 mg Cl2/L; 0,004 kg CO2e/m^3)	**--**	**+0.1 kg/yr**	**+1.7 kg**	Stages: Chlorination (calcium hypochlorite tablet feeder). Sized by the Triton water-quality module
Mains Potable Water Avoided	20.8 m^3/yr (0.00 kg CO2/m^3)	**--**	--	--	Editable potable-water factor; replace with the inventory of the entity that supplies the site

4. 4. CARBON BALANCE, ECO-PAYBACK & METHODOLOGY

Total Embodied Carbon	Annual Pumping	Chlorination / Treatment	Avoided Water Credit	Net Annual Saved	Eco-Payback
+160.4 kg CO2e	**+0.0 kg/yr**	**+0.1 kg/yr (0,048 kg product/yr at 1.50 mg Cl2/L)**	**-0.0 kg/yr**	**+0.1 kg CO2e/yr**	**> 50 years**

Carbon Balance Formulation & Convention: *Official convention: Carbon emitted to atmosphere is POSITIVE (+) in red (environmental impact); carbon avoided/removed from atmosphere is NEGATIVE (-) in green (ecological benefit).*
20-Year Net Balance (kg CO2e) = Embodied CO2 (+) + [Pumping CO2 (+) + Treatment CO2 (+) - Avoided Grid Water (-)] × 20 yrs. Environmental payback represents the time required for operational avoided emissions to offset initial material embodied carbon.

Treatment train considered: Chlorination (calcium hypochlorite tablet feeder).

5. 5. ESG RATING & GREEN BUILDING CERTIFICATIONS

🏛️ LEED v4 / v4.1 (WE)

0 of 8 points (Max: 8 pts)

Scored 0/8 pts (Below the LEED threshold (min. 25%))

WEc1 (Ext.): 0/2 pts | WEc2 (Int.): 0/6 pts

🏛️ LEED v5 BD+C (WEc2)

2 of 8 points

Options 2 to 6

Option 2: 0 | 4: 0 | 6: 2 | 1: 0

🌎 EDGE (IFC)

Water bar: 9.5%

Short by 10.5 p.p.

No points: EDGE measures percentage against a local base case. WEM14 catchment area: 150 m²

💧 GRI 303 & Water Positive

20.8 m³/yr preserved

Ratio: 0.10x

417 m³ over 20 yrs

🌿 ESG Executive Synthesis & Decarbonization

The project reaches an overall potable-water substitution of 9.5% (indoor: 9.5%, outdoor: 0.0%), saving 21 m³/yr of treated water (417 m³ accumulated over 20 years). Under LEED v4 WE (maximum of 8 points), the project scored 0 of 8 points. For EDGE, which awards no points but measures a percentage against a baseline case, the water bar reaches 9.5% — below the 20% threshold required in each of the three bars. The rainwater harvesting system plays a vital role in urban water security and climate resilience, contributing directly to Scope 3 GHG mitigation.

⚠️ Technical Responsibility & Methodological Notice:

This environmental report was generated by Triton following the applicable technical standards and the ISO 14040/14044 guidelines. Emission factors represent screening engineering scenarios and must be validated by the responsible environmental professional against supplier EPDs and the parameters of the local grid and water supply.

Rainfall: weather data from Open-Meteo.com (CC BY 4.0), based on the ERA5 reanalysis of the Copernicus Climate Change Service (Hersbach et al., 2020).